



ARCHITECTURE 架构

Hyrum's Law 海伦姆定律

With a sufficient number of API users, all observable behaviors of your system will be depended on by somebody.
当 API 用户数量足够多时，你系统的所有可观察行为都将受到某些用户的依赖。



ARCHITECTURE 架构

Gall's Law 加尔定律

A complex system that works is invariably found to have evolved from a simple system that worked.
一个运转良好的复杂系统，无一例外都是由一个运转良好的简单系统演变而来。



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The Law of Leaky Abstractions 抽象泄漏定律

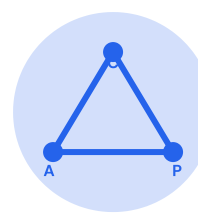
All non-trivial abstractions, to some degree, are leaky.
所有非凡的抽象在某种程度上都是有缺陷的。



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Tesler's Law (Conservation of Complexity) 泰斯勒定律（复杂性守恒定律）

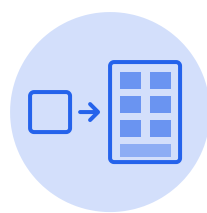
Every application has an inherent amount of irreducible complexity that can only be shifted, not eliminated.
每个应用程序都存在一定程度的固有复杂性，这种复杂性只能转移，而不能消除。



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CAP Theorem CAP 定理

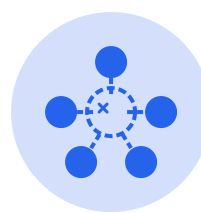
A distributed system can guarantee only two of: consistency, availability, and partition tolerance.
分布式系统只能保证以下三者中的两项：一致性、可用性和分区容错性。



ARCHITECTURE 架构

Second-System Effect 第二系统效应

Small, successful systems tend to be followed by overengineered, bloated replacements.
小型、成功的系统之后，往往会出现过度设计、臃肿的替代系统。



ARCHITECTURE 架构

Fallacies of Distributed Computing 分布式计算的谬误

A set of eight false assumptions that new distributed system designers often make.
新手分布式系统设计师常犯的几个错误假设。



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Law of Unintended Consequences 意料之外的后果定律

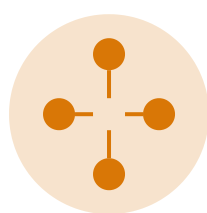
Whenever you change a complex system, expect surprise.
当你改变一个复杂的系统时，要做好应对外部情况的准备。



ARCHITECTURE 架构

Zawinski's Law 扎温斯基定律

Every program attempts to expand until it can read mail.
每个程序都会尝试扩展，直到能够读取邮件为止。



TEAMS 团队

Conway's Law 康威定律

Organizations design systems that mirror their own communication structure.
组织会设计与自身沟通结构相呼应的系统。



TEAMS 团队

Brooks's Law 布鲁克斯定律

Adding manpower to a late software project makes it later.
给一个已经延期的软件项目增加人手只会让项目延期。



TEAMS 团队

Dunbar's Number 邓巴数

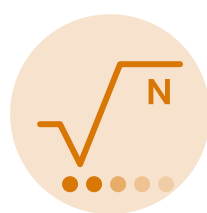
There is a cognitive limit of about 150 stable relationships one person can maintain.
一个人能够维持的稳定关系数量存在认知极限，大约为150个。



TEAMS 团队

The Ringelmann Effect 林格尔曼效应

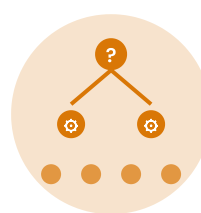
Individual productivity decreases as group size increases.
随着群体规模的增大，个人生产力会下降。



TEAMS 团队

Price's Law 普莱斯定律

The square root of the total number of participants does 50% of the work.
参与者总数的平方根完成了 50% 的工作。



TEAMS 团队

Putt's Law 普特定律

Those who understand technology don't manage it, and those who manage it don't understand it.
懂技术的人不擅长管理技术，而擅长管理技术的人却不了解技术。



TEAMS 团队

Peter Principle 彼得原理

In a hierarchy, every employee tends to rise to their level of incompetence.
在等级制度中，每个员工往往会晋升到自己不胜任的级别。



TEAMS 团队

Bus Factor 巴士因素

The minimum number of team members whose loss would put the project in serious trouble.
团队中至少需要失去多少名成员才会使项目陷入严重困境。



TEAMS 团队

Dilbert Principle 迪尔伯特原理

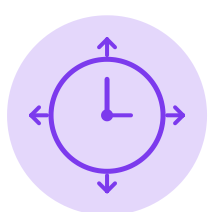
Companies tend to promote incompetent employees to management to limit the damage they can do.
公司往往会提拔不称职的员工担任管理层，以减少他们可能造成的损失。



PLANNING 规划

Premature Optimization (Knuth's Optimization Principle) 过早优化（克努特优化原理）

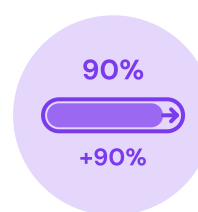
Premature optimization is the root of all evil.
过早优化是万恶之源。



PLANNING 规划

Parkinson's Law 帕金森定律

Work expands to fill the time available for its completion.
工作量会不断增加，直至填满所有可用的完成时间。



PLANNING 规划

The Ninety-Ninety Rule 九九十法则

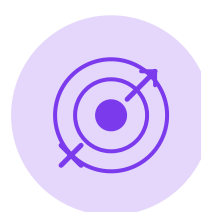
The first 90% of the code accounts for the first 90% of development time; the remaining 10% accounts for the other 90%.
前 90% 的代码占用了前 90% 的开发时间；剩下的 10% 占用了另外 90% 的开发时间。



PLANNING 规划

Hofstadter's Law 霍夫斯塔特定律

It always takes longer than you expect, even when you take into account Hofstadter's Law.
即使考虑到霍夫斯塔特定律，实际花费的时间也总是比你预期的要长。



PLANNING 规划

Goodhart's Law 古德哈特定律

When a measure becomes a target, it ceases to be a good measure.
当一项指标成为目标时，它就不再是一项好的指标了。



PLANNING 规划

Gilb's Law 吉尔布定律

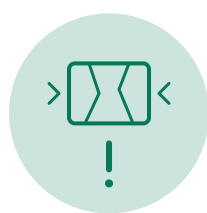
Anything you need to quantify can be measured in some way better than not measuring it.
任何需要量化的事物，总有一种方法可以测量，而且这种方法比不测量要好。



QUALITY 质量

The Boy Scout Rule 童子军规则

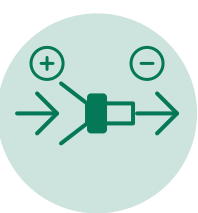
Leave the code better than you found it.
让代码比你最初发现它时更好。



QUALITY 质量

Murphy's Law / Sod's Law 墨菲定律/索德定律

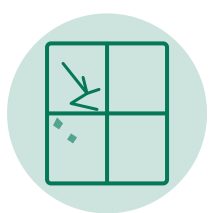
Anything that can go wrong will go wrong.
凡事皆有可能出错，而且一定会出错。



QUALITY 质量

Postel's Law 波斯特尔定律

Be conservative in what you do, be liberal in what you accept from others.
做事要保守，接受别人的东西要开明。



QUALITY 质量

Broken Windows Theory 破窗理论

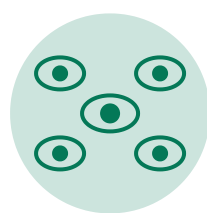
Don't leave broken windows (bad designs, wrong decisions, or poor code) unrepaired.
不要让破损的窗户（糟糕的设计、错误的决策或低劣的代码）得不到修复。



QUALITY 质量

Technical Debt 技术债务

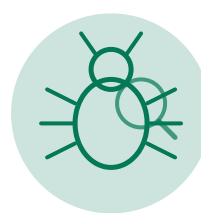
Technical Debt is everything that slows you down when developing software.
技术债务是指在软件开发过程中拖慢我们速度的一切因素。



QUALITY 质量

Linus's Law 莱纳斯定律

Given enough eyeballs, all bugs are shallow.
只要有足够的人关注，所有的虫子都会变得浅薄。



QUALITY 质量

Kernighan's Law 克尼根定律

Debugging is twice as hard as writing the code in the first place.
调试的难度是编写代码的两倍。



QUALITY 质量

Testing Pyramid 测试金字塔

A project should have many fast unit tests, fewer integration tests, and only a small number of UI tests.
一个项目应该有很多快速的单元测试，较少的集成测试，以及少量的 UI 测试。



QUALITY 质量

Pesticide Paradox 农药悖论

Repeatedly running the same tests becomes less effective over time.
反复进行相同的测试，随着时间的推移，效果会越来越差。



QUALITY 质量

Lehman's Laws of Software Evolution 雷曼软件演化定律

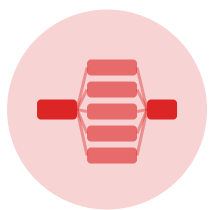
Software that reflects the real world must evolve, and that evolution has predictable limits.
反映现实世界的软件必须不断发展，而这种发展是有可见的局限性的。



QUALITY 质量

Sturgeon's Law 斯特金定律

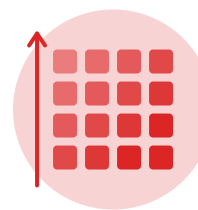
90% of everything is crap.
90%的东西都是垃圾。



SCALE 规模

Amdahl's Law 阿姆达尔定律

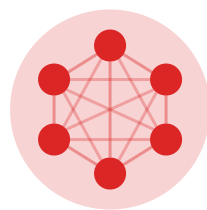
The speedup from parallelization is limited by the fraction of work that cannot be parallelized.
并行化带来的加速效果受限于无法并行化的工作比例。



SCALE 规模

Gustafson's Law 古斯塔夫森定律

It is possible to achieve significant speedup in parallel processing by increasing the problem size.
通过增加问题规模，可以显著提高并行处理的速度。



SCALE 规模

Metcalfe's Law 梅特卡夫定律

The value of a network is proportional to the square of the number of users.
网络的价值与用户数量的平方成正比。



DESIGN 设计

YAGNI (You Aren't Gonna Need It) YAGNI（你不需要它）

Don't add functionality until it is necessary.
不到必要的时候不要添加功能。



DESIGN 设计

DRY (Don't Repeat Yourself) 请勿重复

Every piece of knowledge must have a single, unambiguous, authoritative representation.
每—项知识都必须有单一、明确、权威的表述。



DESIGN 设计

KISS (Keep It Simple, Stupid) KISS（保持简单，笨蛋）

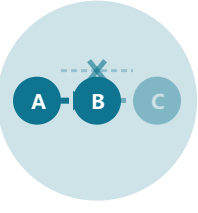
Designs and systems should be as simple as possible.
设计和系统应该尽可能简单。



DESIGN 设计

SOLID Principles SOLID 原则

Five main guidelines that enhance software design, making code more maintainable and scalable.
五项主要指导原则，可增强软件设计，使代码更易于维护和扩展。



DESIGN 设计

Law of Demeter 德墨忒尔法则

An object should only interact with its immediate friends, not strangers.
物体应该只与它的近亲互动，而不应该与陌生人互动。



DESIGN 设计

Principle of Least Astonishment 最小惊讶原则

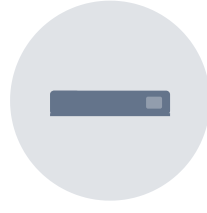
Software and interfaces should behave in a way that least surprises users and other developers.
软件 and 界面的运行方式应该尽量避免让用户和其他开发者感到意外。



DECISIONS 决策

Dunning-Kruger Effect 邓宁-克鲁格效应

The less you know about something, the more confident you tend to be.
对某件事了解得越少，往往就越自信。



DECISIONS 决策

Hanlon's Razor 汉隆剃刀

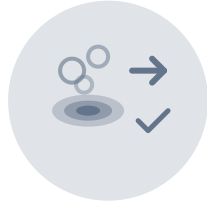
Never attribute to malice that which is adequately explained by stupidity or carelessness.
永远不要把可以用愚蠢或粗心大意解释的事情归咎于恶意。



DECISIONS 决策

Occam's Razor 奥卡姆剃刀

The simplest explanation is often the most accurate one.
最简单的解释往往是最准确的。



DECISIONS 决策

Sunk Cost Fallacy 沉没成本谬误

Sticking with a choice because you've invested time or energy in it, even when walking away helps you.
即使放弃对你更有利，但因为你已经投入了时间和精力，所以仍然坚持自己的选择。



DECISIONS 决策

The Map Is Not the Territory 地图并非疆域本身

Our representations of reality are not the same as reality itself.
我们对现实的认知与现实本身并不相同。



DECISIONS 决策

Confirmation Bias 确认偏差

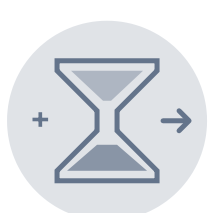
A tendency to favor information that supports our existing beliefs or ideas.
倾向于选择支持我们现有信念或想法的信息。



DECISIONS 决策

The Hype Cycle & Amara's Law 炒作周期与阿马拉定律

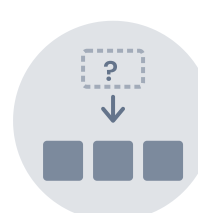
We tend to overestimate the effect of a technology in the short run and underestimate the impact in the long run.
我们往往高估一项技术在短期内的影响，而低估其在长期内的影响。



DECISIONS 决策

The Lindy Effect 林迪效应

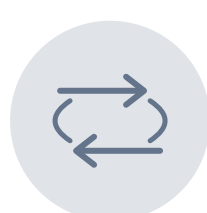
The longer something has been in use, the more likely it is to continue being used.
某物使用时间越长，就越有可能继续被使用。



DECISIONS 决策

First Principles Thinking 第一性原理思维

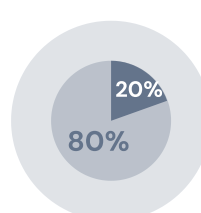
Breaking a complex problem into its most basic blocks and then building up from there.
将复杂问题分解成最基本的部分，然后从这些基本部分入手解决问题。



DECISIONS 决策

Inversion 倒置

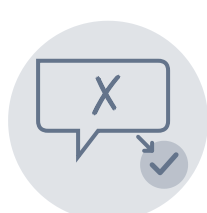
Solving a problem by considering the opposite outcome and working backward from it.
通过考虑相反的结果并从中反向推导来解决问。



DECISIONS 决策

Pareto Principle (80/20 Rule) 帕累托法则（80/20 法则）

80% of the problems result from 20% of the causes.
80% 的问题是由 20% 的原因造成的。



DECISIONS 决策

Cunningham's Law 坎宁安定律

The best way to get the correct answer on the Internet is not to ask a question, it's to post the wrong answer.
在互联网上获得正确答案的最佳方法不是提问，而是发布错误答案。